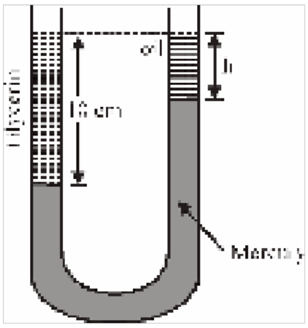
PHYSICS

SECTION-1

1) Solid expand on heating because :-

(1) Kinetic energy of atom increases
(2) Potential energy of atom increases
(3) Total energy of atom increases
(4) The potential energy curve is asymmetric about equilibrium distance

2) A vertical U-tube of uniform inner cross section contains mercury in both sides of its arms. A glycerin (density = 1.3/gcm^3) column of length 10 cm is introduced into one of its arms. Oil of density 0.8 g/cm^3 is poured into the other arm until the upper surfaces of the oil and glycerin are in the same horizontal level. Find the length of the oil column. Density of mercury = 13.6 g/cm^3 :-



(1) 10.4 cm
(2) 8.2 cm
(3) 7.2 cm
(4) 9.6 cm

3) Solid ball of 200 gm at 20°C is dropped in an equal amount of water at 80°C. The resulting temperature is 60°C. This means the specific heat of solid is :-

(1) one fourth of water
(2) one half of water
(3) twice of water
(4) four times of water

4) 1 mole of a gas with $\gamma = 7/5$ is mixed with 1 mole of a gas with $\gamma = 5/3$, then the value of γ for the resulting mixture is-

(1) 7/5
(2) 2/5

(3) 24/16

(4) 12/7

5)

Match the column-I with column-II

Column-I		Column-II	
(A)	Isobaric process	(i)	Constant pressure
(B)	Isothermal process	(ii)	No heat exchange
(C)	Adiabatic process	(iii)	Constant internal energy
(D)	Isochoric process	(iv)	Work done is zero

(1) A–(i) ; B–(ii) ; C–(iii) ; D–(iv)

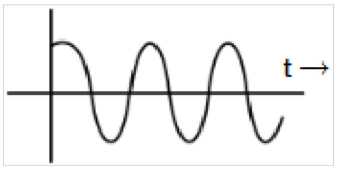
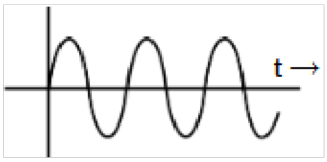
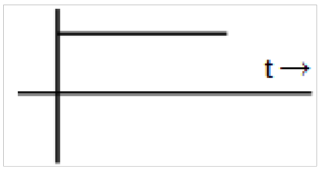
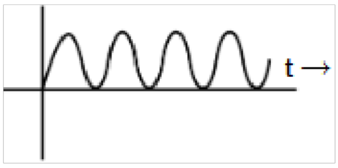
(2) A–(i) ; B–(iii) ; C–(ii) ; D–(iv)

(3) A–(i) ; B–(ii) ; C–(iv) ; D–(iii)

(4) A–(ii) ; B–(i) ; C–(iii) ; D–(iv)

6)

The equation of SHM is given by y = A sin ωt. Match column-I with column-II and select correct option :-

	Column-I		Column-II
(A)	Position vs time graph	(p)	
(B)	Velocity vs time graph	(q)	
(C)	Potential energy vs time graph	(r)	
(D)	Total energy vs time graph	(s)	

(1) A→p, B→q, C→r, D→s

(2) A→q, B→p, C→s, D→r

(3) A→r, B→q, C→p, D→s

(4) A→s, B→r, C→q, D→p

7) **Assertion :** When a solid sphere is placed in the fluid under high pressure, then it is compressed uniformly on all sides.
Reason : The force applied by fluids acts in perpendicular direction at each point of surface.

(1) Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Assertion and Reason are true and Reason is not correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

8) A cube of ice has a iron piece frozen inside. The cube floats in a beaker filled with water, when the ice melts then, the level of water in beaker. :-

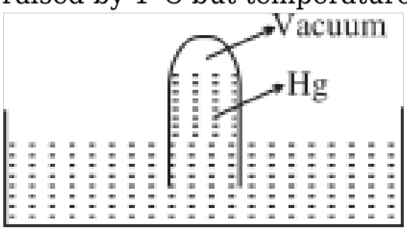
(1) remains same
(2) level decreases
(3) level increases
(4) first increases then decreases

9) A cubical block of steel of each side equal to l is floating on mercury in a vessel. The densities of steel and mercury are ρ_s and ρ_m. The height of the block above the mercury level is given by :

(1) $l\left(1+\frac{\rho s}{\rho m}\right)$
(2) $l\left(1-\frac{\rho s}{\rho m}\right)$
(3) $l\left(1+\frac{\rho m}{\rho s}\right)$
(4) $l\left(1-\frac{\rho m}{\rho s}\right)$

10) ***Assertion* (A)** :- Mean free path of gas molecules, varies inversely as density of the gas.
***Reason* (R)**:- Mean free path of gas molecules is defined as the average distance travelled by a molecule between two successive collisions.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is incorrect but (R) is correct
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

11) A barometer made of a very narrow tube (see fig.) is placed at normal temperature and pressure. The coefficient of volume expansion of Hg is 0.000018/°C and that of the tube is negligible. The temperature of mercury in the barometer is now raised by 1°C but temperature of atmosphere does not change. Then the mercury height in the tube 

(1) Increases
(2) Decreases
(3) Remains same
(4) Cannot say

12) The force acting on a 4gm mass in the energy region $U = 8x^2$ dynes at x = -2 cm is :

(1) 8 dyne
(2) 4 dyne
(3) 16 dyne
(4) 32 dyne

13) If the degrees of freedom of a gas molecule be f, then the ratio of two specific heats C_p/C_v is given by :-

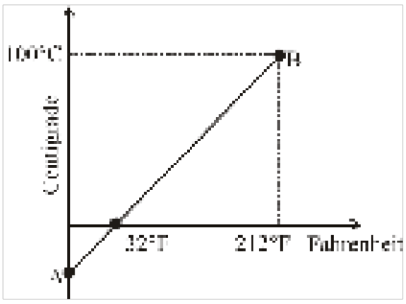
(1) $\frac{2}{f}+1$
(2) $1-\frac{2}{f}$
(3) $1+\frac{1}{f}$
(4) $1-\frac{1}{f}$

14) For a particle executing SHM according to x = A sin ωt, match the columns.

List-I		List-II	
(i)	Minimum time to go at extreme	(P)	$\frac{\pi}{\omega}$
(ii)	Minimum time to come again at mean position	(Q)	$t=\frac{\pi}{2\omega}$
(iii)	Minimum time when velocity is half of maximum value	(R)	$t=\frac{\pi}{3\omega}$
(iv)	Minimum time to cover a displacement from x = 0 to $x=\frac{A}{\sqrt{2}}$	(S)	$t=\frac{\pi}{4\omega}$

(1) i-P; ii-Q; iii-S; iv-R
(2) i-R; ii-S; iii-Q; iv-P
(3) i-S; ii-R; iii-P; iv-Q
(4) i-Q; ii-P; iii-R; iv-S

15) The graph AB shown in figure is a plot of temperature of a body in degree Celsius and degree

Fahrenheit. Then 

(1) slope of line AB is 9/5

(2) slope of line AB is 5/9

(3) slope of line AB is 1/9

(4) slope of line AB is 3/9

16) If for hydrogen $s_p - s_v = a$ and for oxygen $s_p - s_v = b$, where s_p and s_v refer to gram specific heats at constant pressure and at constant volume then

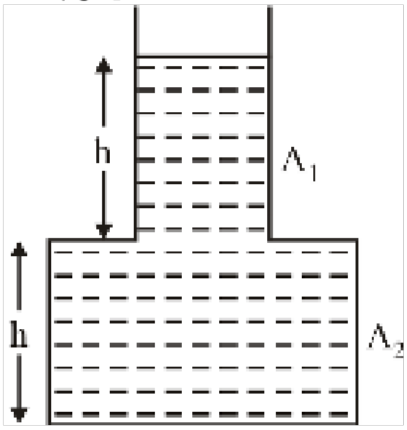
(1) a = b

(2) a = 16b

(3) b = 16a

(4) a and b are not related

17) A vessel shown in the figure has two sections of area of cross-section A_1 and A_2. A liquid of density ρ fill both the sections upto a height h in each case. Neglect atmospheric pressure then
(a) The pressure at the base of the vessel is 2 hρg
(b) The force exerted by the liquid on the base of the vessel is $2h\rho gA_2$
(c) The force exerted by the liquid on the base of the vessel is $h\rho g (A_1 + A_2)$
(d) Weight of the liquid is $2h\rho gA_2$

Correct statements are- 

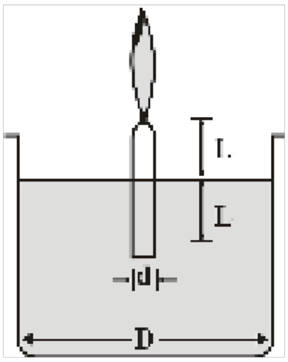
(1) a, b, d

(2) b, c, d

(3) a, b

(4) a, c

18) A candle of diameter d is floating on a liquid in a cylindrical container of diameter D (D>>d) as shown in figure. If it is burning at the rate of 2 cm/hour then the top of the candle will



(1) remain at the same height
(2) fall at the rate of 1 cm/hour
(3) fall at the rate of 2 cm/hour
(4) go up the rate of 1 cm/hour

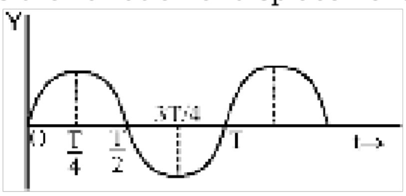
19) If transmission power of a surface is $\frac{1}{9}$, reflective power is $\frac{1}{6}$, what is absorptive power?

(1) $\frac{18}{13}$
(2) $\frac{13}{18}$
(3) $\frac{3}{15}$
(4) $\frac{15}{3}$

20) Two spherical black bodies of radii r_1 and r_2 and with surface temperature T_1 and T_2 respectively radiate the same power. Then the ratio of r_1 and r_2 will be :-

(1) $\left(\frac{T_2}{T_1}\right)^2$
(2) $\left(\frac{T_2}{T_1}\right)^4$
(3) $\left(\frac{T_1}{T_2}\right)^2$
(4) $\left(\frac{T_1}{T_2}\right)^4$

21) The graph shows the variation of displacement of a particle executing SHM with time. We say from this graph is :-



(1) The force is zero at time 3T/4
(2) The velocity is maximum at time T/2

(3) The acceleration is maximum at time T

(4) The P.E. is equal to half of total energy at time T/2

22) A particle performing SHM on the y axis according to equation y = A + B sinωt. Its amplitude is :

(1) A

(2) B

(3) A + B

(4) $\sqrt{A^2 + B^2}$

23) The coefficient of volume expansion of glycerin is 49×10^{-5} k^{-1}. The fractional change in the density on 30°C rise in temperature is :-

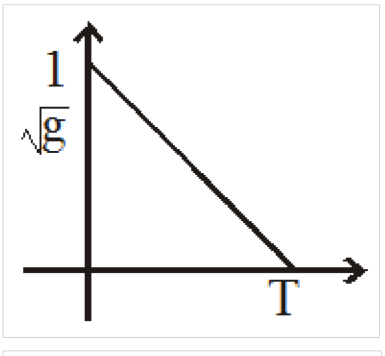
(1) 1.47×10^{-2}

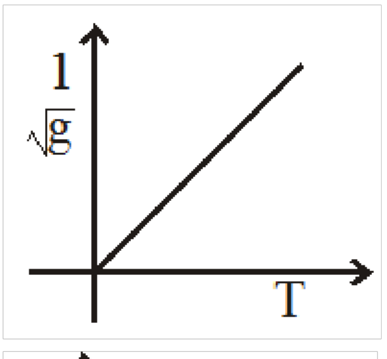
(2) 1.47×10^{-3}

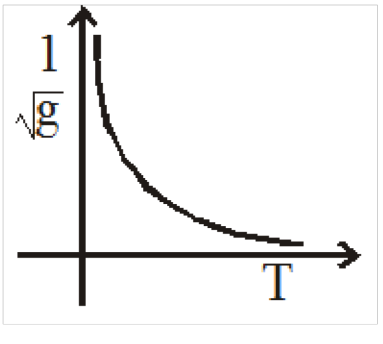
(3) 1.47×10^{-1}

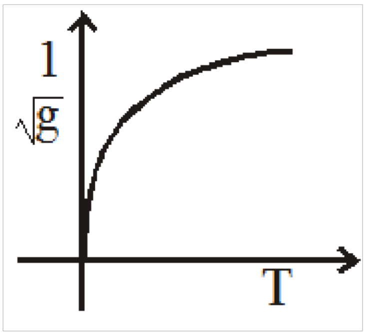
(4) 1.47×10^{-4}

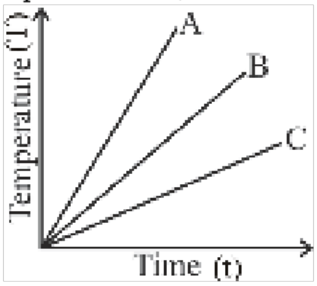
24) The correct curve between $\frac{1}{\sqrt{g}}$ and T (T is time period of simple pendulum) is:-

(1) 

(2) 

(3) 

(4) 

25) Which of the substance A, B or C of same mass has the highest specific heat, if heat is given to A, B and C at same rate ? The temperature v/s time graph is shown :- 

(1) A
(2) B
(3) C
(4) All have equal specific heat

26) The graph between rms speed and temperature for a gas is:-

(1) Parabola
(2) Straight line
(3) Hyperbola
(4) Circle

27) A diatomic gas has pressure P and volume V. It is now compressed adiabatically to $\frac{1}{32}$ times the original volume. The final pressure is

(1) 32 P
(2) 64 P
(3) 128 P
(4) 256 P

28) The flow speeds of air on the lower and upper surfaces of the wing of an aeroplane are v and $\sqrt{5}$ v respectively. The density of air is ρ and surface area of wing is A. The dynamic lift on the wing is :

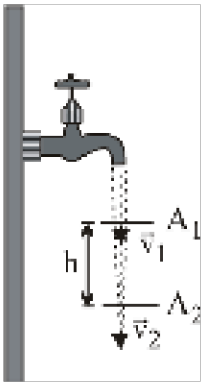
(1) $\rho v^2 A$
(2) $\sqrt{2}\rho v^2 A$
(3) $(1/2)\ \rho v^2 A$
(4) $2\rho v^2 A$

29)

A body cools from 80°C to 60°C in 2 minutes. The time it takes to cool from 60°C to 40°C, when the temperature of the surroundings is 10°C, will be :

(1) 4 min
(2) 3 min
(3) 6 min
(4) 2 min

30) A steady stream of water falls straight down from a pipe. Assume the flow is incompressible ; the flow is similar to figure. How does the pressure in the water vary with height in the falling stream ?



(1) The pressure in the water is higher at lower points in the stream
(2) The pressure in the water is lower at lower points in the stream
(3) The pressure of water is maximum at the middle of the stream
(4) The pressure in the water is the same at all point in the stream

31) A large tank is filled with water to a height H. A small hole is made at the base of the tank. It takes T_1 time to decrease the height of water to H/η, $(\eta > 1)$ and it takes T_2 time to take out the rest of water. If $T_1 = T_2$, then the value of η is :

(1) 2
(2) 3
(3) 4
(4) $2\sqrt{2}$

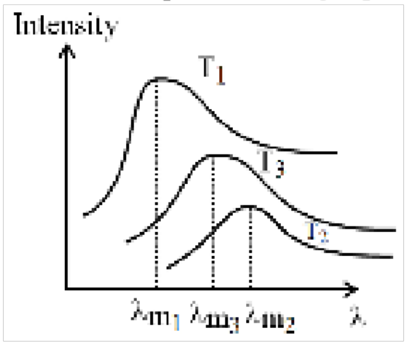
32) 100 g of ice at 0°c is mixed with 100 g of water at 80°C. The final temperature of mixture will be:-

(1) 0°C
(2) 140°C
(3) 80°C
(4) < 0°C

33) 64 identical water droplets falling in air with velocity 5 cm/s. If these droplets coalase then find the new constant velocity :

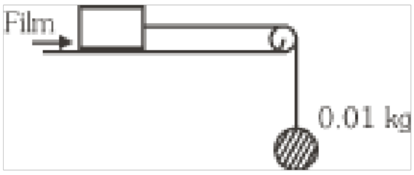
(1) 40 cm/s
(2) 20 cm/s
(3) 160 cm/s
(4) 80 cm/s

34) The plots of intensity versus wavelength for three black bodies at temperatures T_1, T_2 and T_3 respectively are as shown. Their temperature are such that :-



(1) $T_1 > T_2 > T_3$
(2) $T_1 > T_3 > T_2$
(3) $T_2 > T_3 > T_1$
(4) $T_3 > T_2 > T_1$

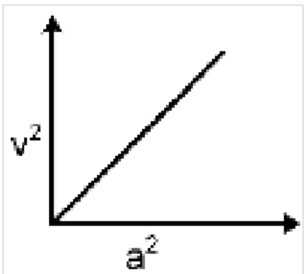
35) A metal block of area 0.10 m^2 is connected to a 0.010 kg mass via a string that passes over an ideal pulley (considered massless & frictionless). A liquid with a film thickness of 0.30 mm is placed between the block and the table. When released, the block moves to the right with a constant speed of 0.085 m/s. Find the coefficient of viscosity of the liquid. ($g = 9.8\ m/s^2$)

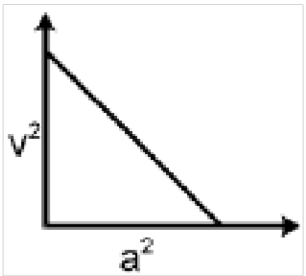


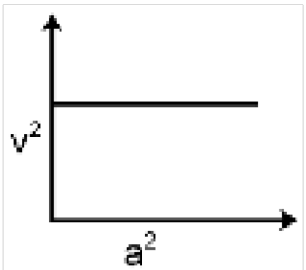
(1) 6×10^{-2} Pa.s
(2) 3.45×10^{-5} Pa.s
(3) 3.45×10^{-3} Pa.s
(4) 3.45×10^{-6} Pa.s

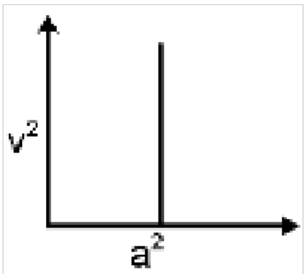
SECTION-2

1) A mass M is performing linear simple harmonic motion, then correct graph for v^2 and a^2 (where a is acceleration and v is linear velocity):-

(1) 

(2) 

(3) 

(4) 

2) A container, whose bottom has round holes with diameter 0.1 mm is filled with water. The maximum height in cm upto which water can be filled without leakage will be what? Surface tension = 75×10^{-3} N/m and g = 10 m/s^2 :

(1) 20 cm

(2) 40 cm

(3) 30 cm

(4) 60 cm

3) Suppose ideal gas equation follows VP^3 = constant. Initial temperature and volume of the gas are T and V respectively. If gas expands to 27V, then its temperature will become :

(1) T

(2) 9T

(3) 27T

(4) T/9

4) A large number of liquid drops each of radius r coalesce to from a single drop of radius R. The energy released in the process is converted into kinetic energy of the drop so formed. The speed of the big drop is (given, surface tension of liquid T, density ρ)

(1) $\sqrt{\frac{T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

(2) $\sqrt{\frac{2T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

(3) $\sqrt{\frac{4T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

(4) $\sqrt{\frac{6T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

5) A soap bubble of radius r is blown upto form a bubble of radius 2r under isothermal conditions. If T is the surface tension of soap solution, the energy spent in the blowing, is :-

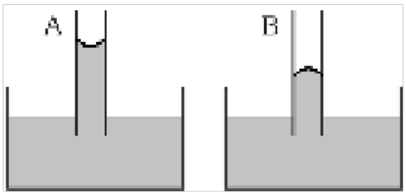
(1) $3\pi Tr^2$ joule

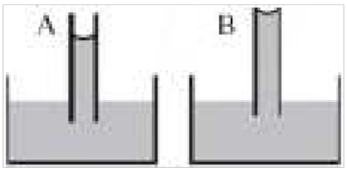
(2) $6\pi Tr^2$ joule

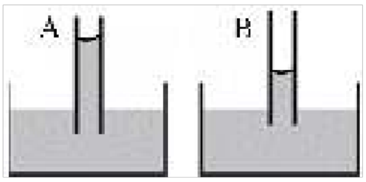
(3) $12\pi Tr^2$ joule

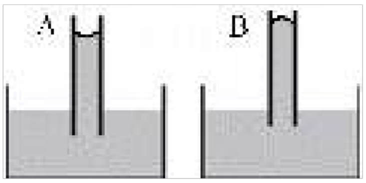
(4) $24\pi Tr^2$ joule

6) A capillary tube (1) is dipped in water. Another identical tube (2) is dipped in a soap -water solution. Which of the following shows the relative nature of the liquid columns in the two tubes?

(1) 

(2) 

(3) 

(4) 

7) Which of the following is constant during SHM:

(1) Velocity

(2) Acceleration

(3) Total energy

(4) Phase

8) In a simple harmonic motion, the ratio of K.E. of the particle at mean position to the point when distance is half of amplitude will be:

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $\frac{4}{3}$

(4) $\frac{3}{2}$

9) The temperature gradient in a rod of 0.5 m length is 80°C/m. If the temperature of hotter end of the rod is 30°C, then the temperature of the cooler end is :-

(1) 40°C

(2) –10°C

(3) 10°C

(4) 0°C

10) **Statement-1 :** If a spring of spring constant 'K' is cut into 'n' equal parts then spring constant of each part is 'nK'
Statement-2 : Spring constant 'K' is directly proportional to natural length of spring.

(1) Both statement 1 and statement 2 are true

(2) Both statement 1 and statement 2 are false

(3) Statement 1 is true but statement 2 is false

(4) Statement 1 is false but statement 2 is true

11) A vessel contains 110 g of water. The heat capacity of the vessel is equal to 10g of water. The initial temperature of water in vessel is 10°C. If 220 g of hot water at 72°C is poured in the vessel, the final temperature neglecting radiation loss, will be :

(1) 70°C

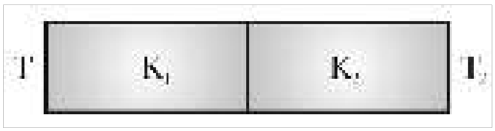
(2) 80°C

(3) 60°C

(4) 50°C

12)

If two metallic plates of equal thicknesses and thermal conductivities K_1 and K_2 are put together face to face and a common plate is constructed, then the equivalent thermal conductivity of this plate will be :



(1) $\frac{K_1 K_2}{K_1 + K_2}$

(2) $\frac{2K_1 K_2}{K_1 + K_2}$

(3) $\dfrac{\left[K_1^2 K_2^2\right]^{3/2}}{K_1 K_2}$

(4) $\dfrac{\left[K_1^2 + K_2^2\right]^{3/2}}{2K_1 K_2}$

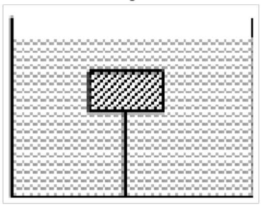
13) Three liquids of densities ρ_1, ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$), having the same value of surface tension T, rise to the same height in three identical capillaries. The angles of contact θ_1, θ_2 and θ_3 obey:-

(1) $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$

(2) $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$

(3) $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geqslant 0$

(4) $0 \leqslant \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$

14) Tension in string is T_0 when system is at rest. What will be the tension in string if system has upward acceleration a? 

(1) T_0

(2) $2T_0$

(3) $T_0\left(\frac{a}{g}\right)$

(4) $T_0\left(1+\frac{a}{g}\right)$

15) Water is entering in a cylindrical container open from top at the rate of α. Container has an orifice of area A at base. Find maximum height upto which water can be filled :

(1) $\frac{\alpha}{2gA}$

(2) $\frac{\alpha^2}{2gA^2}$

(3) $\frac{\alpha^2}{2gA}$

(4) $\frac{\alpha}{2gA^2}$

CHEMISTRY

SECTION-1

1) The standard molar heat of formation of ethane, CO_2 and water(ℓ) are respectively –21.1, –94.1 and –68.3 kCal. The standard molar heat of combustion of ethane will be

(1) –372 kCal
(2) –162 kCal
(3) –240 kCal
(4) –183.5 kCal

2)

	Column-I		Column-II
(A)	$C_{diamond} + O_{2(g)} \rightarrow CO_{2(g)}$	(i)	Enthalpy of formation
(B)	$HCOOH_{(aq)} + NaOH_{(aq)} \rightarrow HCOONa_{(aq)} + H_2O_{(l)}$	(ii)	Enthalpy of hydration
(C)	$\frac{1}{2}H_{2(g)} + \frac{1}{2}Cl_{2(g)} \rightarrow HCl_{(g)}$	(iii)	Enthalpy of combustion
(D)	$CuSO_{4(s)} + 5H_2O_{(l)} \rightarrow CuSO_4.5H_2O_{(s)}$	(iv)	Heat liberated by neutralisation will be less then 13.7 Kcal

(1) A–iii, B–iv, C–i, D–ii
(2) A–ii, B–iv, C–i, D–iii
(3) A–iii, B–ii, C–iv, D–i
(4) A–iii, B–iv, C–ii, D–i

3) If $H_{2\,(g)} \rightarrow 2H_{\,(g)}$; ΔH = 104 kcal, then heat of atomisation of H_2 (g) is :-

(1) 52 kcal
(2) 104 kcal
(3) 208 kcal
(4) 5.2 kcal

4) The bond dissociation energy of gaseous H_2, Cl_2 and HCl are 104, 58 and 103 kcal mol^{-1} respectively. The enthalpy of formation for HCl gas will be

(1) –44.0 kcal
(2) –22.0 kcal
(3) 22.0 kcal
(4) 44.0 kcal

5) The enthalpy of formation of ammonia is –46.0 kJ mol^{-1}. The enthalpy change for the reaction $2NH_3(g) \rightarrow 2N_2(g) + 3H_2(g)$ is

(1) 46.0 kJ mol^{-1}
(2) 92.0 kJ mol^{-1}

(3) –23.0 kJ mol^{-1}

(4) –92.0 kJ mol^{-1}

6) The heat of neutralisation is the highest for the reaction between :-

(1) $NH_4OH + CH_3COOH$

(2) $HNO_3 + NH_4OH$

(3) $NaOH + CH_3COOH$

(4) $HCl + NaOH$

7) If the enthalpies of formation of Al_2O_3 and Cr_2O_3 are –1596 kJ and –1134 kJ respectively, then the value of ΔH for the reaction;
$2Al + Cr_2O_3 \rightarrow 2Cr + Al_2O_3$ will be :

(1) –462 kJ

(2) –136 kJ

(3) –2530 kJ

(4) +2530 kJ

8) Which of the following is path function?

(1) Heat

(2) Internal energy

(3) Enthalpy

(4) Entropy

9) Given these reactions :
$A \rightarrow 2B$ $\quad \Delta H = -40$ kJ
$B \rightarrow C$ $\quad \Delta H = -50$kJ
$2C \rightarrow D$ $\quad \Delta H = -20$ kJ
Calculate ΔH for the reaction : $D + A \rightarrow 4C$

(1) –100 kJ

(2) –60 kJ

(3) –120 kJ

(4) 100 kJ

10) The heat change for the reaction,
$C(s) + 2S(s) \rightarrow CS_2(\ell)$ is known as :-
(Graphite)

(1) heat of transition of C

(2) heat of fusion of $CS_2(\ell)$

(3) heat of formation of $CS_2(\ell)$

(4) heat of vaporisation of $CS_2(\ell)$

11) Energy required to dissociate 4 g of gaseous hydrogen into free gaseous atoms is 208 kcal at 25°C. The bond energy of H—H bond will be:

(1) 1.04 cal mol^{-1}

(2) 10.4 kcal mol^{-1}

(3) 104 kcal mol^{-1}

(4) 1040 kcal mol^{-1}

12) A system absorb 600J of heat and work equivalent to 300J on its surroundings. The change in internal energy is

(1) 300 J

(2) 400 J

(3) 500 J

(4) 600 J

13) Which is correct for an adiabatic process ?

(1) $\Delta T = o$

(2) $q = 0$

(3) $\Delta V = 0$

(4) $\Delta P = 0$

14) 2 mole of an ideal gas expanded isothermally and reversiblly and reversibly from 1 L to 10 L at 300 K. What is the enthalpy change?

(1) 4.98 kJ

(2) 11.47 kJ

(3) –11.7 kJ

(4) 0 kJ

15) A process has $\Delta H = 200$ J mol^{-1} and $\Delta S = 40$ JK^{-1} mol^{-1}. Out of the values given below, choose the minimum temperature above which the process will be spontaneous

(1) 20 K

(2) 4 K

(3) 5 K

(4) 12 K

16) The standard entropies of H_2(g), I_2(g) and HI(g) are 130, 116.7 and 206.3 J K^{-1} mol^{-1} respectively. The change in standard entropy in the reaction :
H_2 (g) + I_2 (g) → 2HI (g) is :

(1) 185.6 J K^{-1} mol^{-1}

(2) 170.5 J K^{-1} mol^{-1}

(3) 158.9 J K^{-1} mol^{-1}

(4) 165.9 J K^{-1} mol^{-1}

17) Which has the highest entropy per mol of the substance :-

(1) H_2 at 25°C and 1 atm

(2) H_2 at STP

(3) H_2 at 100 K and 1 atm

(4) H_2 at 0 K and 1 atm

18) During which of the following processes, does entropy decrease?
(A) Dissolution of NaCl in water
(B) Freezing of water to ice at –10°C
(C) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
(D) Adsorption of CO(g) on lead surface

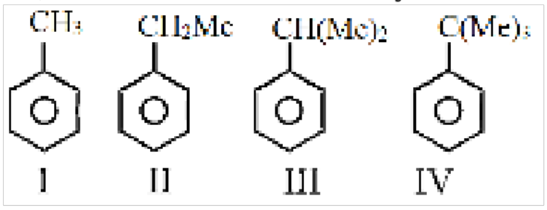
(1) (A), (B), (C) and (D) only

(2) (A), (B) and (C) only

(3) (A) and (B) only

(4) (B), (C) and (D) only

19) Correct order of reactivity of following for ESR (Electrophilic substitution reaction) is :-

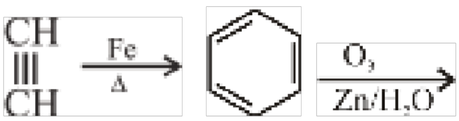


(1) IV > III > II > I

(2) IV > II > III > I

(3) I > II > III > IV

(4) I > IV > II > III

20) 

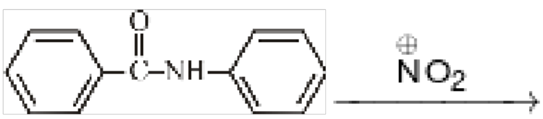
Y is :-

(1) $\begin{array}{l} CH_2-OH \\ | \\ CH_2-OH \end{array}$

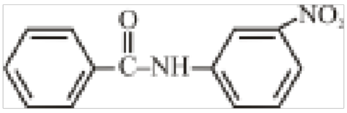
(2) CH_3-CH_2-OH

(3) CH_3-COOH

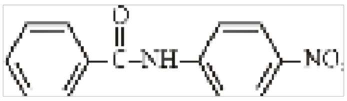
(4) $\begin{array}{l} CH=O \\ | \\ CH=O \end{array}$

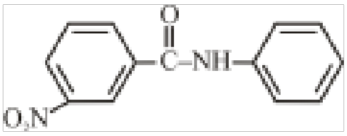
21) 

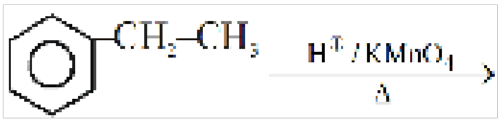
Main product is:-

(1) 

(2) 

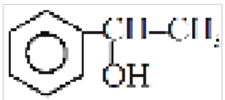
(3) 

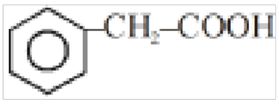
(4) 

22)  ?

(1) 

(2) 

(3) 

(4) 

23) Which of the following group is ortho and para directing :-

(i) CH_3– (ii) –NO_2

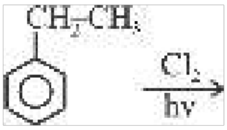
(iii) –NH_2 (iv) –OH (v) –COOH

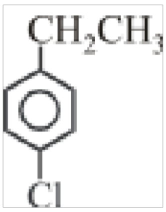
(1) i, ii, iii

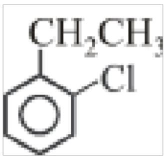
(2) i, iii, iv

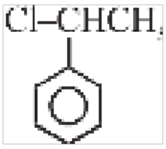
(3) i, iii, v

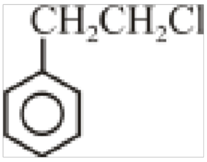
(4) iii, iv, v

24)  Product (major), find product :-

(1) 

(2) 

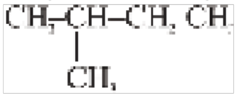
(3) 

(4) 

25) Arrange the following compound in order of their boiling point ?

(A) $CH_3CH_2CH_3$

(B) $CH_3CH_2CH_2CH_3$

(C) 

(D) CH_3-CH_2-CH_2-CH_2-CH_3

(1) D > C > B > A

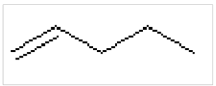
(2) A > B > C > D

(3) C > D > B > A

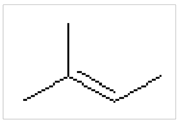
(4) C > B > D > A

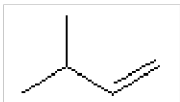
26)

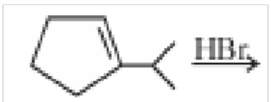
Which of the following Alkene is least stable?

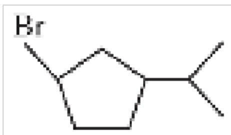
(1) 

(2) 

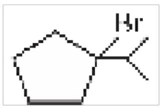
(3) 

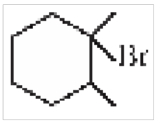
(4) 

27) Which of the following products is not formed in following reaction ? 

(1) 

(2) 

(3) 

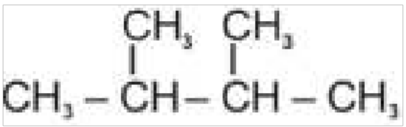
(4) 

28) Match the column:-

Column-I (Reactants)	Column-II (Product)
(a) $CaC_2 + H_2O$	(p) CaC_2
(b) $CaO + C$	(q) H_2
(c) $HC \equiv CH + Na$	(r) CH_4
(d) $CH_3COOH + NaOH + CaO$	(s) C_2H_2

(1) a-q, b-p, c-r, d-s
(2) a-p, b-q, c-s, d-r
(3) a-s, b-p, c-q, d-r
(4) a-p, b-q, c-r, d-s

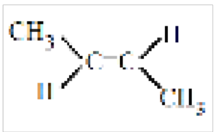
29) Which one is not prepared by Wurtz reaction?

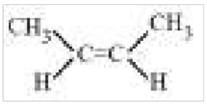
(1) C_2H_6
(2) $n-C_4H_{10}$
(3) CH_4
(4) 

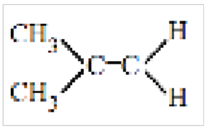
30) In Kharasch effect; reaction follows

(1) Free radical substitution
(2) Electrophilic addition
(3) Free radical addition
(4) Nucleophilic addition

31) The compound which reacts with HBr obeying Markownikov's rule is :-

(1) $CH_2 = CH_2$
(2) 

(3) 

(4) 

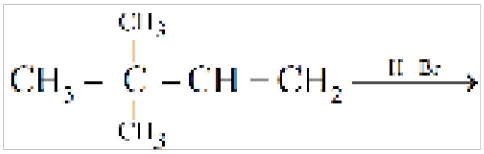
32) Which is chain teminating step in the following :-

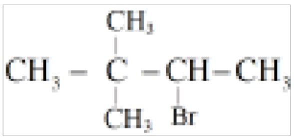
(1) $H_2 \rightarrow H\bullet + \underset{\bullet}{H}$

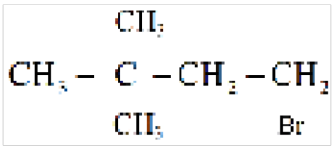
(2) $Br_2 \rightarrow Br\bullet + \underset{\bullet}{Br}$

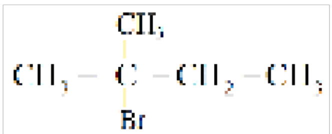
(3) $H\bullet + HBr \rightarrow H_2 + Br\bullet$

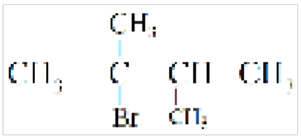
(4) $Br\bullet + Br\bullet \rightarrow Br_2$

33) The major product formed in the given reaction are:- 

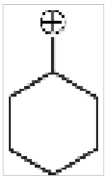
(1) 

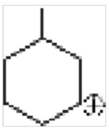
(2) 

(3) 

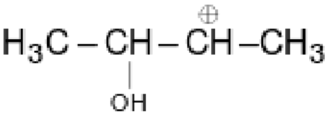
(4) 

34) Which of the following carbocation does not show rearrangement?

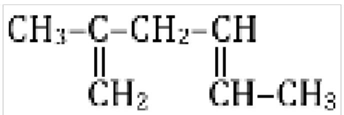
(1) 

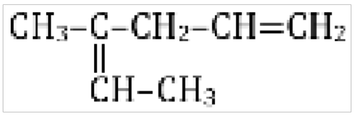
(2) 

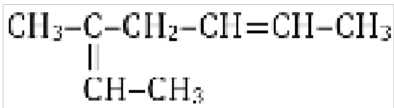
(3) $CH_3-CH_2-NH-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\overset{\oplus}{C}H_2$

(4) 

35) The structure of reactant which gives H-CHO, CH_3-CHO and CH_3-$COCH_2$-CHO on ozonolysis ?

(1) 

(2) 

(3) 

(4) Both (1) and (2)

SECTION-2

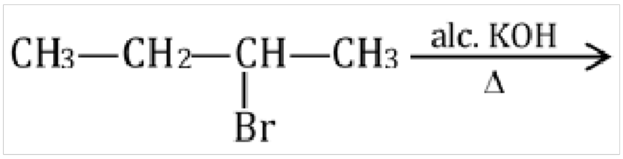
1) Which is best method to obtain methane ?

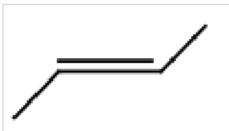
(1) $CH_3-Cl + Na \xrightarrow[\text{ether}]{\text{dry}}$

(2) $CH_3-CH_2-COONa \xrightarrow{\text{electrolysis}}$

(3) $CH_3-Cl + Ph-Cl \xrightarrow[\text{ether}]{\text{dry}}$

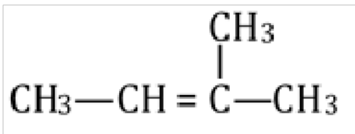
(4) $CH_3-Cl \xrightarrow{Zn/HCl}$

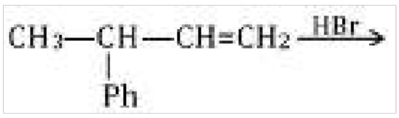
2)  X(major product):-

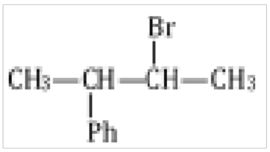
(1) 

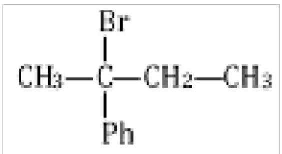
(2) $CH_3-CH=CH_2$

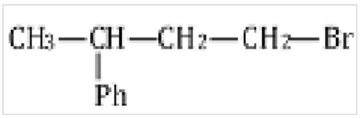
(3) 

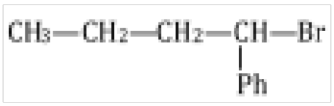
(4) 

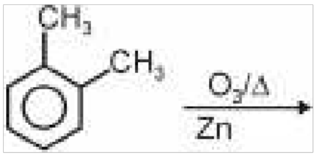
3)  (A) What will be major (A)?

(1) 

(2) 

(3) 

(4) 

4) 

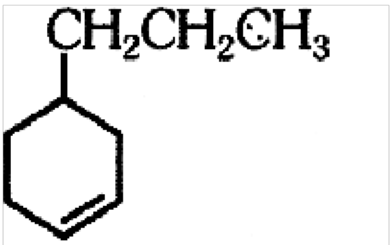
How many different species are found ?

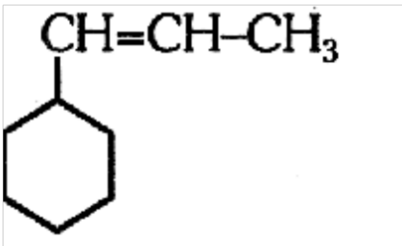
(1) 1
(2) 2
(3) 3
(4) 4

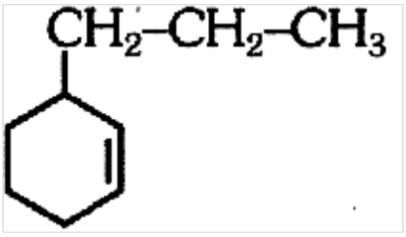
5) **Statement I:** Cis-but-2-ene has less dipole moment than trans-but-2-ene.
Statement II: R–C≡C–R' reacts with H_2 in presence of Na/liq. NH_3 to form cis-alkene.

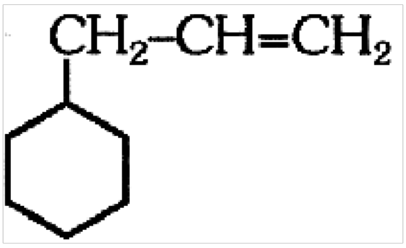
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct
(3) Statement-I and Statement-II are correct.
(4) Both Statement-I and II are incorrect

6) An alkene on ozonolysis gives methanal as one of the product. Its structure is:

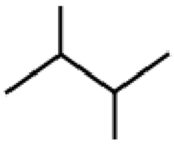
(1) 

(2) 

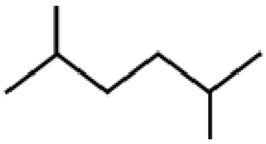
(3) 

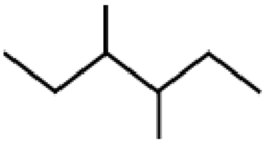
(4) 

7) Which of the following is not suitable to obtain from Wurtz reaction :-

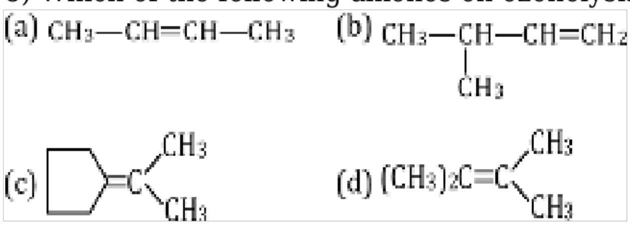
(1) 

(2) 

(3) 

(4) 

8) Which of the following alkenes on ozonolysis give a mixture of ketones only?



(1) a and b

(2) b and c

(3) b and d

(4) c and d

9) Match list I with list II and then select the correct answer from the codes given below the lists :-

List-I		List-II	
(A)	Na/Dry Ethers	(p)	Decarboxylation
(B)	Zn-Hg/HCl	(q)	Clemmenson

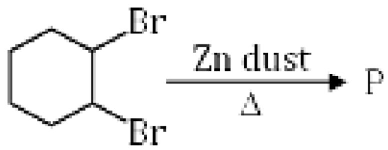
(C)	$O_3/Zn/H_2O$	(r)	Wurtz reaction
(D)	NaOH/CaO	(s)	Ozonolysis

(1) A-r, B-q, C-p, D-s

(2) A-r, B-q, C-s, D-p

(3) A-p, B-q, C-r, D-s

(4) A-s, B-q, C-r, D-p

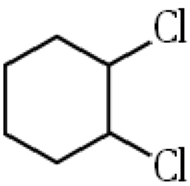
10) 

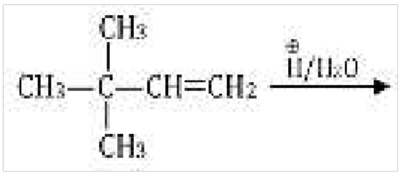
'P' is :-

(1) 

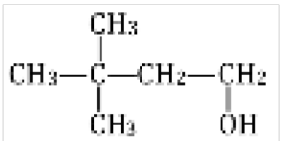
(2) 

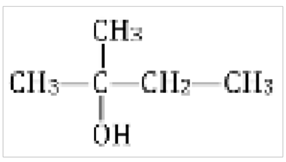
(3) 

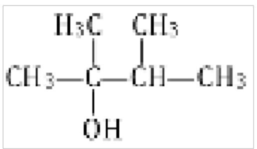
(4) 

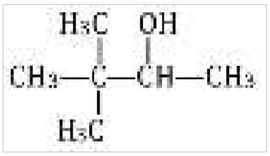
11) 

Major product will be :-

(1) 

(2) 

(3) 

(4) 

12) Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas ?

(1) $C_P + C_V = R$

(2) $C_P - C_V = R$

(3) $C_P = RC_V$

(4) $C_V = RC_P$

13) The correct option for free expansion of an ideal gas under adiabatic condition is :

(1) $q > 0$, $\Delta T > 0$ and $w > 0$

(2) $q = 0$, $\Delta T = 0$ and $w = 0$

(3) $q = 0$, $\Delta T < 0$ and $w > 0$

(4) $q < 0$, $\Delta T = 0$ and $w = 0$

14) For the reaction, $2Cl(g) \rightarrow Cl_2(g)$, the **correct** option is :-

(1) $\Delta_rH < 0$ and $\Delta_rS < 0$

(2) $\Delta_rH > 0$ and $\Delta_rS > 0$

(3) $\Delta_rH > 0$ and $\Delta_rS < 0$

(4) $\Delta_rH < 0$ and $\Delta_rS > 0$

15) In which case change in entropy is negative ?

(1) Evaporation of water

(2) Expansion of a gas at constant temperature

(3) Sublimation of solid to gas

(4) $2H(g) \rightarrow H_2(g)$

BOTANY

SECTION-1

1) In dicots initiation of lateral roots occur from the region of:

(1) Cortex

(2) Endodermis

(3) Cork cambium

(4) Pericycle

2) How many of the tissues in the list given below are included in ground tissue system ?
Cortex, Pericycle, Epidermis, Hypodermis, Pith, Medullary rays, Stomata, Trichomes, Mesophyll, Endodermis

(1) Seven
(2) Six
(3) Five
(4) Eight

3) The pith is small or inconspicuous in :

(1) Monocot root
(2) Dicot root
(3) Dicot stem
(4) Monocot stem

4) **Assertion (A) :** Radial vascular bundles are present in roots.
Reason (R) : Xylem and phloem within a vascular bundle are arranged in an alternate manner on different radii.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) (A) is true but (R) is false
(3) (A) is false but (R) is true
(4) Both (A) and (R) are true and (R) is the correction explanation of (A)

5) Which of the following structure constitute stele:

(1) Pericycle
(2) Vascular bundles
(3) Pith
(4) All of these

6) Casparian strips that are observed on endodermis are made up of :-

(1) Suberin
(2) Lignin
(3) Hemicellulose
(4) Cellulose

7) Which of the following is **not** a part of stomatal apparatus

(1) Stomatal aperture
(2) Guard cells
(3) Subsidiary cells
(4) Complimentary cells

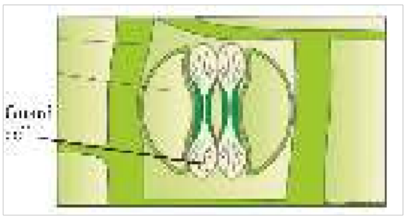
8) T.S. of a material exhibits conjoint and closed vascular bundles scattered in a ground tissue what

should be the material?

(1) Monocot root
(2) Dicot root
(3) Monocot stem
(4) Dicot stem

9) Position of xylem and phloem in leaf are respectively :-

(1) Adaxial and adaxial
(2) Adaxial and abaxial
(3) Abaxial and adaxial
(4) Abaxial and abaxial

10)  Identify the given figure.

(1) Stomata of dicot plant
(2) Stomata of grasses
(3) Stomata of pteridophyte
(4) Stomata of gymnosperm

11) Innermost layer of cortex is called :-

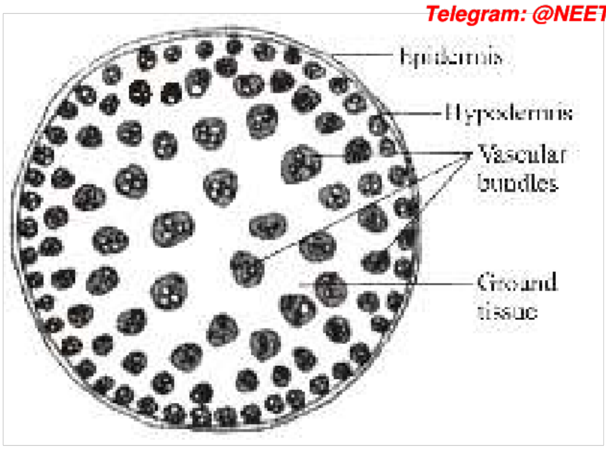
(1) Hypodermis
(2) Endodermis
(3) Pericycle
(4) Pith rays

12) In grasses, the guard cells are :-

(1) Bean-shaped
(2) Star shaped
(3) Cruciform
(4) Dumb-bell shaped

13) In dicot roots parenchymatous cells which lie between the xylem and phloem are called :-

(1) Casparian strips
(2) Passage cells
(3) Conjuctive tissue
(4) Bulliform cells

14) What is **true** for the diagram given below ? 

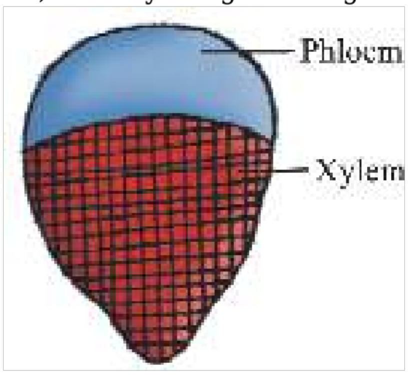
(1) Vascular bundles are present in the form of a ring

(2) Pith is distinct

(3) Hypodermis is collenchymatous

(4) Water-containing cavities are present within the vascular bundles

15) Identify the given diagram :-



(1) Conjoint open type vascular bundle

(2) Conjoint closed type vascular bundle

(3) Radial type vascular bundle

(4) Concentric type vascular bundle

16) **Statement-1 :** In dicot stem, the endodermis is also referred to as the starch sheath.
Statement-2 : The cells of the endodermis are rich in starch grains.

(1) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1.

(2) Statement–1 is false, Statement–2 is true.

(3) Statement–1 is true, Statement–2 is false.

(4) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1.

17) If a plant group is classified on the basis of chromosome number, structure, behaviour etc. then the classification is based on?

(1) Chemotaxonomy

(2) NOR of chromosomes

(3) Cytotaxonomy

(4) All

18) Holdfast, stipe and frond constitutes the plant body in case of

(1) Chlorophyceae

(2) Phaeophyceae

(3) Rhodophyceae

(4) Cyanophyceae

19) Which of the following is a colonial alga ?

(1) *Fucus*

(2) *Ulothrix*

(3) *Volvox*

(4) *Chlorella*

20) Isogamy with non flagellated gametes is seen in:

(1) *Volvox*

(2) *Fucus*

(3) *Ectocarpus*

(4) *Spirogyra*

21) Smallest Angiosperm is–

(1) *Wolffia*

(2) *Sequoia*

(3) *Eucalyptus*

(4) *Cycas*

22) Which of the following is responsible for peat formation?

(1) *Sphagnum*

(2) *Funaria*

(3) *Anthoceros*

(4) *Polytrichum*

23) Floridean starch is found in?

(1) Chlorophyceae

(2) Rhodophyceae

(3) Phaeophyceae

(4) None of the above

24) Which one of the following pair is heterosporous forms?

(1) *Sphagnum* and *Salvinia*
(2) *Lycopodium* and *Marchantia*
(3) *Selaginella* and *Salvinia*
(4) *Dryopteris* and *Adiantum*

25) Which of the following is **not** a moss?

(1) *Polytrichum*
(2) *Sphagnum*
(3) *Funaria*
(4) *Marchantia*

26) Gymnosperms are referred to as "naked seeded plants", because

(1) They lack ovule
(2) Ovules are not enclosed by any ovary wall
(3) They have no seed coat
(4) The embryo is unprotected

27) Strobili or cones are found in :

(1) *Selaginella, Equisetum*
(2) *Equisetum, Azolla*
(3) *Equisetum, Dryopteris*
(4) *Selaginella, Sphagnum*

28) Agar is obtained from :

(1) *Fucus*
(2) *Laminaria*
(3) *Gelidium*
(4) *Chara*

29) Post fertilisation development can be seen in :

(1) *Polysiphonia* and *Porphyra*
(2) *Fucus* and *Gelidium*
(3) *Chara* and *Porphyra*
(4) *Polysiphonia* and *Volvox*

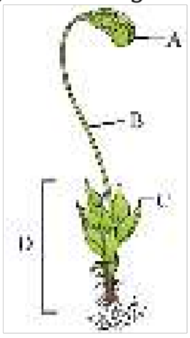
30) Which of the following can be considered as first terrestrial plants with vascular tissues?

(1) Algae
(2) Fungi
(3) Bryophyte
(4) Pteridophyte

31) Gemmae are asexual buds located on the thallli :-

(1) *Dictyota*
(2) *Polytrichum*
(3) *Marchantia*
(4) *Funaria*

32) Given below is the diagram of organism identify the parts labelled A, B, C and D, and select the **right** option about them :-



	Part (A)	Part (B)	Part (C)	Part (D)
1	Sporophyte	Stalk	Leaves	Foot
2	Capsule	Foot	Leaves	Gametophyte
3	Sporophyte	Stalk	Foot	Gametophyte
4	Capsule	Seta	Leaves	Gametophyte

(1) 1
(2) 2
(3) 3
(4) 4

33) Pyrenoid present in algae is made up of :-

(1) Starch only
(2) Starch and protein
(3) Protein only
(4) Any polysaccharide

34) In *Cycas*

(1) Microsporangia and ovule are present in same sporophyte.
(2) Micro and megasporophylls are present in same cone.
(3) Male cone and megasporophylls are borne on different trees.
(4) Male cone and megasporophylls are borne on the same plant.

35) Mark the correct statement w.r.t. the plant given below– 

(1) It is heterosporous
(2) It has seed inside the fruit
(3) It is a moss
(4) It is a fern

SECTION-2

1) Casparian strips are found on the -

(1) Tangential and radial walls of pericycle cells of stem
(2) Tangential and radial walls of endodermal cells of dicot roots
(3) Tangential and radial walls of hypodermal cells of stem
(4) Only radial walls of endodermal cells of monocot roots.

2) Mark the **correct** sequence of layers found in root anatomy from outside to inside:

(1) Epidermis, cortex, endodermis, pericycle
(2) Cortex, Epidermis, pericycle, endodermis
(3) Epidermis, pericycle, endodermis, cortex
(4) Cortex, Epidermis, endodermis, pericycle.

3) This diagram represents.

(1) A dicotyledon plant
(2) A pteridophyte
(3) A Thallophyte
(4) A Bryophyte

4) All tissues except __(A)__ and __(B)__ constitute the ground tissue :-

(1) A = Epidermis, B = Hypodermis
(2) A = Hypodermis, B = Pericycle
(3) A = Endodermis, B = Pericycle
(4) A = Epidermis, B = Vascular bundles

5) In grasses, certain adaxial epidermal cells along the veins modify themselves into large empty colourless cells. These cells are called :-

(1) Subsidary cells
(2) Bulliform cells
(3) Cork cells
(4) Guard cells

6)

Which of the following statement(s) is/are **true**?
(a) Usually trichomes are multicellular
(b) Root hairs are unicellular

(c) The trichomes help in preventing water loss due to transpiration
(1) Only a
(2) Only a & b
(3) Only b & c
(4) All a, b & c

7) Which of the following is generally found in hypodermis of monocots stem?

(1) Collenchyma
(2) Aerenchyma
(3) Sclerenchyma
(4) Chlorenchyma

8) All of the following are **true** for leaves of gymnosperms, **except**

(1) Have thick cuticle
(2) Lack stomata
(3) Can be needle like
(4) Are well adapted to withstand extremes of temperature

9) In *Funaria* (moss) spore germinates to produce?

(1) Protonema
(2) Prothallus
(3) Proembryo
(4) Embryo

10) In gymnosperm, sporangia are born on sporophylls which are arranged .... along an axis to form lax or compact strobili or cones.

(1) Oppositely
(2) Radially
(3) In whorl
(4) Spirally

11) A plant shows thallus level of organisation. It shows rhizoids and is haploid. It needs water to complete its life cycle because the male gametes are motile. It may belong to

(1) Pteridophytes
(2) Gymnosperms
(3) Monocots
(4) Bryophytes

12) Match the classes of pteridophyta given in **Column-I** with their respective members given in **Column-II**

	Column-I		**Column-II**
(a)	Psilopsida	(i)	*Selaginella*
(b)	Lycopsida	(ii)	*Adiantum*
(c)	Pteropsida	(iii)	*Psilotum*
(d)	Sphenopsida	(iv)	*Equisetum*

(1) a(iii), b(i), c(iv), d(ii)
(2) a(i), b(iii), c(ii), d(iv)
(3) a(iii), b(i), c(ii), d(iv)
(4) a(i), b(iv), c(iii), d(ii)

13) Algae are divided into various classes on the basis of :-
(A) Pigment
(B) Morphology of thallus
(C) Stored food
(D) Mode of spore formation and fruiting bodies

(1) A, B, C & D
(2) A & B
(3) C & D
(4) A & C

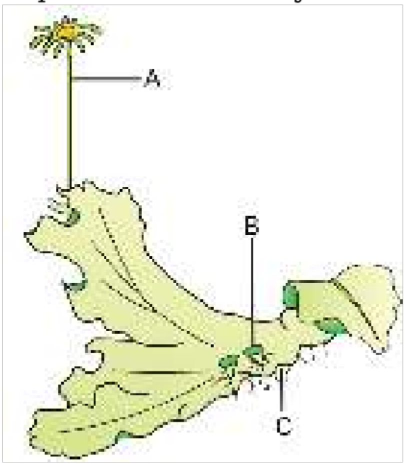
14) "The vegetative cells have a cellulosic wall usually covered on the outside by a gelatinous coating of algin". This statement is **correct** for -

(1) *Chara*
(2) *Volvox*

(3) *Laminaria*

(4) *Polysiphonia*

15) Select the option that **correctly** identifies A, B and C in the given figure of female thallus of *Marchantia*–



(1) A – Gemma cup, B – Antheridiophore , C – Rhizoids

(2) A – Antheridiophore, B – Rhizoids, C – Gemma cup

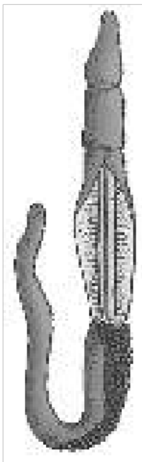
(3) A – Archegoniophore, B – Gemma cup, C – Rhizoids

(4) A–Archegoniophore, B – Rhizoids, C – Gemma cup

ZOOLOGY

SECTION-1

1) Identify the incorrect statement about the given organism :-

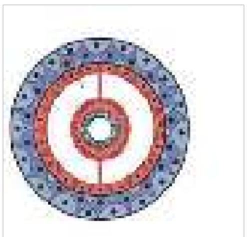


(1) Body in divided into proboscis, collar & trunk

(2) It is commonly called as Acorn worm

(3) Circulatory system is of closed type

(4) External fertilisation is observed.

2) Identify the diagram :- 

(1) Coelomate
(2) Pseudocoelomate
(3) Acoelomate
(4) All of the above

3) Coelenterates and Ctenophores resemble each other in being :-

(1) Monoblastic and acoelomate
(2) Diploblastic and tissue grade of body organisation
(3) Triplobastic and pseudocoelomate
(4) Triploblastic & tissue grade of body organisation

4)  The above structure helps a cnidarian in which of the following given activities ?
(A) *Anchorage* (B) Reproduction
(C) Defense (D) Capturing prey

(1) A, B, C, D
(2) C, D
(3) A, C, D
(4) B, C, D

5) The figure shows four animals (a), (b), (c) and (d). Select the correct answer with respect to common characteristics of two of these animals.


(a) (b) (c) (d)

(1) (c) and (d) have a true coelom
(2) (a) and (d) respire mainly through body wall
(3) (b) and (c) show radial symmetry
(4) (a) and (b) have cnidoblasts for self defence.

6) "Brain coral" is :

(1) *Meandrina*
(2) *Hydra*
(3) *Physalia*
(4) *Obelia*

7)

How many in the given examples of animals are Coelentrates ?

Physalia, Obelia, Planaria, Pennatula, Gorgonia, Pleurobrachia, Meandrina and Nereis
(1) Three
(2) Four
(3) Five
(4) Six

8) An acoelomate animal with bilateral symmetry is :-

(1) Jelly fish
(2) Sponges
(3) Liver fluke
(4) Ctenoplana

9) Which is not the characteristic of phylum porifera ?

(1) Spongocoel
(2) Paragastric cavity
(3) Choanocytes
(4) Tissue level

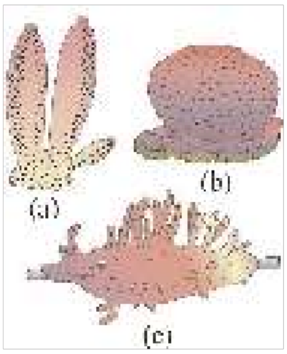
10) Gregarious pest is :-

(1) Limulus
(2) Locusta
(3) Bombyx
(4) Laccifer

11) Which one of the following is a matching pair of a body feature and the animal possessing it?

(1) Ciliary comb plates – Obelia
(2) Setae with parapodia – Nereis
(3) Antennae – Star fish
(4) Mantle – Prawn

12) Given below are the three figures of sponges. Which of the following sponge is found in fresh

water? 

(1) only a
(2) only b
(3) a and b
(4) only c

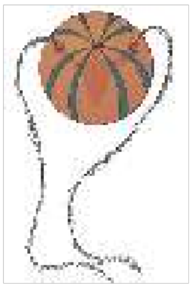
13) Which of the following is correct for medusa ?

(1) Sessile
(2) Cylindrical
(3) Free swimming
(4) Present in Hydra

14) Identify the animal shown below as well as the related type of development and select the right option for the two together : 

	Animal	Type of development
(1)	*Taenia*	Direct
(2)	*Fasciola*	Many larval stages
(3)	*Fasciola*	larva absent
(4)	*Planaria*	larva absent

(1) 1
(2) 2
(3) 3
(4) 4

15) Select the incorrect option for given figure of animal 

(1) Digestion is both extra cellular and intra cellular
(2) Bioluminescence is well-marked
(3) Diploblastic with tissue level of organisation
(4) Fertilisation is internal

16) In which animal excretory organ is proboscis gland ?

(1) *Balanoglossus*
(2) *Dentalium*
(3) *Antedon*
(4) *Pinctada*

17) Which of the following is correct option ?

(1) Laccifer-Locust
(2) Bombyx-Limulus
(3) Prawn-Economically important insect
(4) Limulus-King crab

18) Which of the following option is incorrect for Annelida?

(1) Both longitudinal and circular muscles are found.
(2) Earthworm is bisexual.
(3) Nereis is Monoecious.
(4) Reproduction is sexual.

19) In which of the following option all animals are dioecious

(1) *Fasciola, Ascaris, Taenia*
(2) *Ascaris, Ancylostoma, Wuchereria*
(3) *Ancyclostoma, Fasciola, Ascaris*
(4) *Taenia, Ascaris, Wuchereria*

20) Annelids may be

(1) Aquatic (marine and fresh water), terrestrial
(2) Free living
(3) Parasite

(4) All of the above

21) Consider the following four statements (a-d) for coelentrates and give the answer of question asked below.
(a) Polyp produce medusa asexually
(b) Medusa produce polyp sexually
(c) Polyp are free swimming and cylindrical in shape
(d) Medusa are umbrella shaped and sessile
How many option are correct ?

(1) a, b and c
(2) c and d
(3) b and d
(4) a and b

22) First phylum to have complete digestive tract can be represented by : -

(1) Hydra
(2) Taenia
(3) Ancylostoma
(4) Pheretima

23) The space between body wall and alimentary canal lined by mesoderm is called :-

(1) Acoelom
(2) Pseudocoelom
(3) Coelom
(4) None of these

24) Ostia is present in :-

(1) Porifera
(2) Coelenterata
(3) Annelida
(4) Mollusca

25) Common bath sponge is :-

(1) *Spongilla*
(2) *Euspongia*
(3) *Leucosolenia*
(4) *Sycon*

26) Which one opposes parathormone ?

(1) ADH
(2) Insulin

(3) Thyroxine

(4) Thyrocalcitonin

27) Thymus gland is a lobular structure between _____a_________ behind _____b___ on the _____c_____ side of aorta.

(1) a) lungs b) ribs c) ventral

(2) a) Ribs b) lungs c) dorsal

(3) a) sternum b) ribs c) ventral

(4) a) lungs b) sternum c) ventral

28) Which of the following is correct ?
(A) Pars distalis produces GH,PRL, TSH. ACTH, LH, FSH
(B) Pars intermedia secretes only one hormone called melatonin
(C) Posterior lobe of pituitary is also called as neurohypophysis or pars nervosa
(D) Posterior pituitary, stores and releases two hormones called oxytocin and vasopressin

(1) A,B,C

(2) B,C,D

(3) A,C,D

(4) B and C

29) Parathyroid hormone –

(1) is produced by the thyroid gland

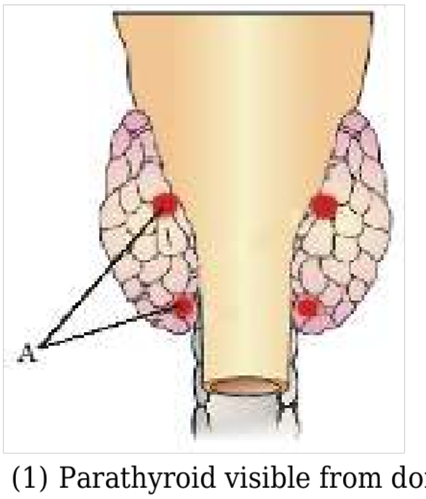
(2) is released when blood calcium levels fall

(3) stimulates osteoblasts to lay down new bone

(4) stimulates calcitonin release.

30)

Identify 'A'



(1) Parathyroid visible from dorsal side

(2) Parathyroid visible from ventral side

(3) Parafollicular cells from dorsal side

(4) Parafollicular cells from ventral side

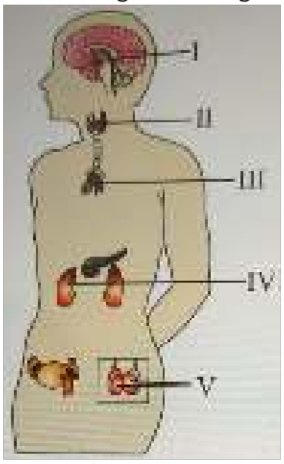
31) Both the lobes of thyroid glands are interconnected with a thin flap of connective tissue called

(1) Ampulla
(2) Infundibulum
(3) Isthmus
(4) Bridge

32) Hormone which is more active than T_4 is :-

(1) T_1
(2) T_2
(3) T_3
(4) T_5

33) The given diagram represents the location of human endocrine glands I, II, III, IV and V.



Which of the following is correctly matched with their secretions?

Hormones	Their secretions
I	Melatonin
II	Thymosin
III	Epinephrine
IV	Aldosterone
V	Testosterone

(1) I, II and III only
(2) I, IV and V only
(3) II, IV and V only
(4) II, III and V only

34) How many hormones from the below list interact with intracellular receptors and regulate chromosome function and gene expression.
GH, Oxytocin, ADH, ANF, Aldosterone, Cortisol, Thyroxin, Epinephrine, Estradiol, GnRH, Progesterone

(1) Four
(2) Five
(3) Six
(4) Seven

35) Vigorous contraction in uterus at the time of parturition regulated by :

(1) Oxytocin
(2) ADH
(3) Adrenaline
(4) ACTH

SECTION-2

1) Which hormone acts on the exocrine pancreas and stimulates secretion of water and bicarbonate ions ?

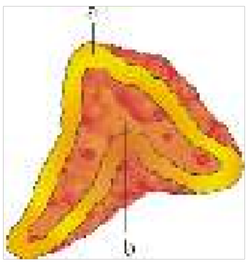
(1) Secretin
(2) Cholecystokinin
(3) Gastric inhibitory peptide
(4) Both (1) and (2)

2) The thyroid gland is composed of:-

(1) Thyroid follicles
(2) Cortical tissues
(3) Stromal tissues
(4) Both 1 and 3

3) Which of the following organ perform dual functions as a primary sex organ as well as an endocrine gland ?

(1) Testis and ovary
(2) Pancreas, testis and ovary
(3) Adrenal cortex, ovary and testis
(4) Penis and vagina

4) Recogine the figure and find out the correct matching :- 

(1) a-adrenal gland, b-kidney

(2) a-kidney, b-adrenal gland

(3) a-adrenal medulla, b-adrenal cortex

(4) a-adrenal cortex, b-adrenal medulla

5) GnRH, a hypothalamic hormone, needed in reproduction, acts on:

(1) anterior pituitary gland and stimulates secretion of LH and FSH.

(2) posterior pituitary gland and stimulates secretion of oxytocin and FSH.

(3) posterior pituitary gland and stimulates secretion of LH and relaxin.

(4) anterior pituitary gland and stimulates secretion of LH and oxytocin

6) Hormones are :-

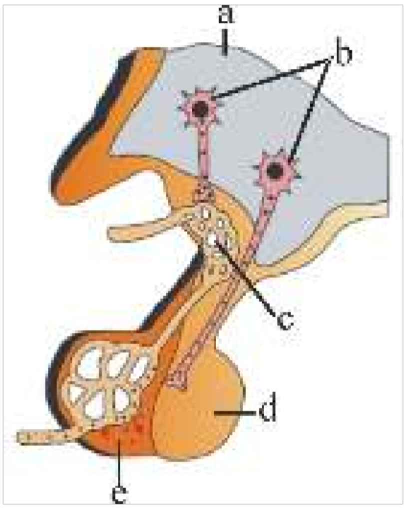
(1) Intra cellular, nutritive substance

(2) Intercellular, Non nutritive substance

(3) Maximum secreted always

(4) Species specific, Non nutritive substance

7) Recognise the figure and find out the correct matching :



(1) a-pituitary, b-portal circulation, c-pars intermedia, d-hypothalamus, e-sella tursica

(2) a-hypothalamus, b-hypothalamic neurons, c-portal circulation, e-posterior pituitary

(3) a-hypothalamus, b-potral circulation, c-hypothalamic neurons, d-posterior pituitary

(4) a-hypothalamus, b-hypothalamic neurons, c-portal circulation, d-posterior pituitary, e-anterior pituitary

8) Diabetes insipidus is due to :-

(1) Hyper secretion of ADH

(2) Hypo secretion of ADH

(3) Hyper secretion of insulin
(4) Hypo secretion of insulin

9) Milk ejection hormone is :-

(1) Prolactin
(2) Adrenaline
(3) Oxytocin
(4) Vasopressin

10) CCK stimulate the secretion of _____ :-

(1) Gastric juice
(2) Bile juice
(3) Intestinal juice
(4) HCl

11) Stunted growth, mental retardation and deaf-mutism in growing baby is due to

(1) Hyperthyroidism during pregnancy
(2) Hyposecretion of thyroxine in mother during pregnancy
(3) Abnormal secretion of oxytocin and prostaglandins during pregnancy
(4) Abnormal secretion of dopamine, inhibits oxytocin and PRL in pregnancy

12) The islets of Langerhans are found in

(1) Stomach
(2) Pancreas
(3) Liver
(4) Alimentary canal

13) A person is suffering from high blood pressure. Which hormone decreases his blood pressure by vasodilation?

(1) A steroid hormone secreted from adrenal medulla
(2) A peptide hormone secreted from atrial wall of heart
(3) A protein hormone secreted from pineal gland
(4) A fatty acid derivative hormone secreted from gonads

14) Hormone that stimulates synthesis and secretion of thyroid hormones :-

(1) T_3
(2) T_4
(3) TSH
(4) ACTH

15) Which of the following is an amino acid derived hormone ?

(1) Epinephrine

(2) Aldosterone

(3) Cortisol

(4) All of these

ANSWER KEYS

PHYSICS

SECTION-1

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	2	3	2	2	1	2	2	1	1	4	1	4	2	2	3	2	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
A.	2	2	1	2	3	1	3	4	2	4	3	1	4	2	3					

SECTION-2

Q.	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	2	3	2	4	4	3	3	3	2	3	4	2	4	4	2

CHEMISTRY

SECTION-1

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	1	1	2	2	2	4	1	1	3	3	3	1	2	4	3	4	1	4	3	4
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85					
A.	3	1	2	3	1	4	1	3	3	3	4	4	4	2	4					

SECTION-2

Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	4	1	2	3	4	4	2	4	2	3	3	2	2	1	4

BOTANY

SECTION-1

Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	1	2	4	4	1	4	3	2	2	2	4	3	4	2	4	3	2	3	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135					
A.	1	1	2	3	4	2	1	3	1	4	3	4	2	3	1					

SECTION-2

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	1	4	2	4	3	2	1	4	4	3	4	3	3

ZOOLOGY

SECTION-1

Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**3**	**1**	**2**	**3**	**1**	**1**	**3**	**3**	**4**	**2**	**2**	**4**	**3**	**2**	**4**	**1**	**4**	**3**	**2**	**4**
Q.	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185					
A.	**4**	**3**	**3**	**1**	**2**	**4**	**4**	**3**	**2**	**1**	**3**	**3**	**2**	**2**	**1**					

SECTION-2

Q.	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
A.	**1**	**4**	**1**	**4**	**1**	**2**	**4**	**2**	**3**	**2**	**2**	**2**	**2**	**3**	**1**

SOLUTIONS

PHYSICS

2)

$P_0 + P_g g(10) = P_0 + P_{oil}\, gh + P_{Hg}\, g(10–h)$

6) $y = A \sin \omega t \Rightarrow$ y-t graph = sin x type graph

$v = \frac{dy}{dt} A\omega \cos \omega t \Rightarrow$ v-t graph = cos x type graph

$PE = \frac{1}{2} Ky^2 = \frac{1}{2} KA^2 \sin^2 \omega t$

$\Rightarrow$ PE – t graph = $\sin^2$ x type graph

$TE = \frac{1}{2} KA^2$ = constant

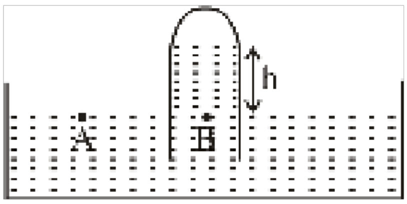
$\Rightarrow$ TE is independent of time

7) If a solid sphere placed in the fluid under high pressure, then it is compressed uniformly on all sides. The force applied by the fluids acts in perpendicular direction at each point of the surface and the body is said to be under hydraulic compression. This leads to decrease in its volume without any change of its geometrical shape.

Therefore, Assertion and Reason are correct and reason is the correct explanation of assertion.

11) Atmosphere is kept unaffected so its presence does not change

Also in barometer



$\rho_A = \rho_B$

$\rho_{atm} = h\, \rho_{Hg}\, g$

$h = \frac{\rho_{atm}}{\rho_{Hg} g} \rightarrow h \propto \frac{1}{\rho_{Hg}}$

Now on heating density of mercury decreases so h increases.

13) $\gamma = \frac{C_p}{C_v} = \frac{\left[\left(\frac{f+2}{2}\right) R\right]}{\left[\frac{fR}{2}\right]} = \frac{f+2}{f}$

$\gamma = \frac{f}{f} + \frac{2}{f} \Rightarrow \gamma = \frac{2}{f} + 1$

15) Slope of line AB

$= \frac{\Delta C}{\Delta F} = \frac{100-0}{212-32} = \frac{100}{180} = \frac{5}{9}$

16)

C = Ms

C: Molar heat capacity

s: Specific heat capacity

M: Molecualr mass

For H_2 : $s_P - s_V = a$

$M_{H_2}(s_P - s_V) = M_{H_2}a$

$C_P - C_V = 2a = R$ ...(i)

For O_2: $s_P - s_V = b$

$M_{O_2}(s_P - s_V) = M_{O_2}b$

$C_P - C_V = 32\ b = R$ ...(ii)

$2a = 32\ b$

$a = 16\ b$

19) t + r + a = 1

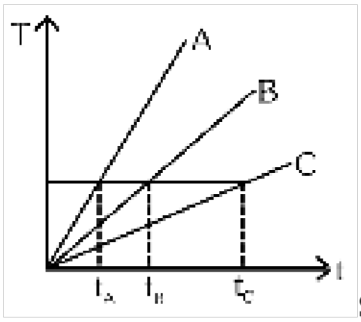
$a = 1 - \left(\frac{1}{9} + \frac{1}{6}\right) = \frac{13}{18}$

20) $\left(\frac{Q}{t}\right)_1 = \left(\frac{Q}{t}\right)_2$

$\sigma 4\pi r_1^2 T_1^4 = \sigma 4\pi r_2^2 T_2^4$

$\frac{r_1}{r_2} = \left(\frac{T_2}{\pi}\right)^2$

23) $\frac{\Delta\rho}{\rho_0} = Y\ \Delta\theta = 49 \times 10^{-5} \times 30 = 1.47 \times 10^{-2}$

25) 

Substance having more specific heat taken longer time to get heated to a higher temperature and longer time to get cooled. If we draw a line parallel to the time axis then it cut the graph at three different points. Corresponding points on the time axis shows that $t_C > t_B > t_A \Rightarrow C_C > C_B > C_A$

26) $V_{rms} \propto \sqrt{T}$

$V^2 \propto T$ (Its the relation of parabola)

28) $F = \Delta PA = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right) A$

$F = \frac{1}{2}\rho\left[\left(\sqrt{5}v\right)^2 - v^2\right] A = 2\rho v^2 A$

29)

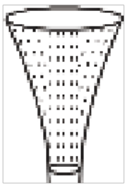
use $\frac{-dT}{dt} = K(T - T_0)$ (T = average temp.)

$\frac{20}{2} = K[70 - 10] \Rightarrow K = \frac{1}{6}$

$\frac{20}{t} = K[50 - 10] \Rightarrow \frac{20}{t} = \frac{40}{6}$

t = 3 min

30) Pressure is atmospheric



31) $T \mu \left(\sqrt{H_1} - \sqrt{H_2}\right)$

$\frac{T_1}{T_2} = \frac{\left(\sqrt{H} - \sqrt{\frac{H}{n}}\right)}{\left(\sqrt{\frac{H}{n}} - 0\right)}$

 $T_1 = T_2 \Rightarrow \frac{T_1}{T_2} = 1$

$\Rightarrow \sqrt{\frac{H}{n}} = \sqrt{H} - \sqrt{\frac{H}{n}}$

$\Rightarrow 2\sqrt{\frac{H}{n}} = \sqrt{H}$

$\Rightarrow n = 4$

32) Heat available = $ms\Delta T = 100 \times 1 \times 80 = 8000$ cal

heat required to melt whole ice = $mL = 100 \times 80 = 8000$ cal

$(\Delta Q)_{req} = (\Delta Q)_{\text{Heat available}}$

Temp of mixture = 0°C

33) $V_{final} = n^{2/3}V = (64)^{2/3} \times 5\text{cm/s} = 16 \times 5\text{cm/s}$

34) According to Wien's law $\lambda_m \propto \frac{1}{T}$ and from the figure $(\lambda m)_1 < (\lambda m)_3 < (\lambda m)_2$ therefore $T_1 > T_3 > T_2$.

35)

f = $\eta A \frac{dv}{dy} \Rightarrow \eta = \frac{0.01 \times 10 \times 0.3 \times 10^{-3}}{0.1 \times 0.085} = 3.45 \times 10^{-3} \text{Pa.s}$

38) VP^3 = constant

$V\left(\frac{nRT}{V}\right)^3$ = constant

$T^3 \propto V^2$

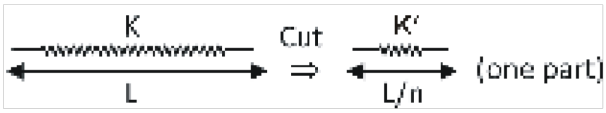
$\frac{T_1}{T_2} = \left(\frac{V_1}{V_2}\right)^{2/3} = \left(\frac{V}{27V}\right)^{2/3} = \frac{1}{9}$

$T_2 = 9T_1 = 9T$

43) $\frac{KE_{mean}}{KE_x} = \frac{\frac{1}{2}kA^2}{\frac{1}{2}k\left[A^2 - \frac{A^2}{\ell}\right]}$

$= \frac{4}{3}$

44) $\frac{30-\theta}{0.5} = 80 \Rightarrow 30-\theta = 40 \Rightarrow \theta = -10^{\circ}C$

45) 

Using KL = constant ⇒ KL = $K'\left(\frac{L}{n}\right)$

⇒ K′ = nK

46) Let final temperature of water be θ

Heat taken = Heat given

⇒ 110 × 1(θ – 10) + 10 (θ – 10)

= 220 × 1(72 – q)

⇒ θ = 48.8°C = 50°C

48) h = $\frac{2T\cos\theta}{\rho g r}$

As r, h, T are same, $\frac{\cos\theta}{\rho}$ = constant

$\Rightarrow \frac{\cos\theta_1}{\rho_1} = \frac{\cos\theta_2}{\rho_2} = \frac{\cos\theta_3}{\rho_3}$

As $\rho_1 > \rho_2 > \rho_3$

$\Rightarrow \cos\theta_1 > \cos\theta_2 > \cos\theta_3 \Rightarrow \theta_1 < \theta_2 < \theta_3$

As water rises so q must be acute

So, $0 \le \theta_1 < \theta_2 < \theta_3 < \pi/2$

49) Initially, $F_B = T_0 + mg$

$\Rightarrow T_0 + F_B - mg$

$\Rightarrow T_0 = V(\sigma - \rho)\, g$ ......(1)

when lift accelerates.

$T = V(\sigma - \rho)(g + a)$ ......(2)

$\Rightarrow T = T_0\left(\frac{g+a}{g}\right) = T_0\left(1 + \frac{a}{g}\right)$

CHEMISTRY

51)

$C_2H_{6(g)} + \frac{7}{2}O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(l)}$; $\Delta H_{Comb.} = ?$

$\Delta H = \Sigma(\Delta E_f)_P - \Sigma(\Delta E_f)_R$

$\Delta H = [2 \times (-94.1) + 3 \times (-68.3)] - [1 \times (-21.1) + 0]$

$\Delta H_{Comb.} = -372$ KCal

52)

Read notes

53) $H_{2(g)} \rightarrow 2H_{(g)}$ $\Delta H_{atomisation}$

56) SA + SB

$\Delta H_N = -13.7$ kcal/gm. Equiv.

57)

The given reaction is :

$2Al(s) + Cr_2O_3(s) \rightarrow Al_2O_3(s) + 2Cr(s)$

$\Delta H = [\Delta_f H^o_{Al_2O_3} + 2\Delta_f H^o_{Cr}] - \left[2\Delta_f H^o_{Al} + \Delta_f H^o_{Cr_2O_3}\right]$

$= [-1596 + 2 \times 0] - [2 \times 0 + (-1134)]$

$= -462$ kJ

59) $\Delta H = +20 + 40 - 50 \times 2$

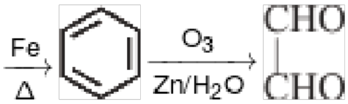
61)

$H_2(g) \rightarrow 2H(g)$
For 4g = 2mol H_2 ; E = 208 kcal
∴ for 1 mol H_2 ; B.E. = $\frac{208}{2}$ = 104 kcal

65)

$\Delta G = \Delta H - T\Delta S$
The process will be spontaneous, when
ΔG = –ve, i.e. $|T\Delta S| > |\Delta H|$
Given : $\Delta H = 200\ J\ mol^{-1}$
and $\Delta S = 40\ JK^{-1}\ mol^{-1}$
$\Rightarrow T > \frac{|\Delta H|}{|\Delta S|} = \frac{200}{40} = 5K$
So, the minimum temperature for spontaneity of the process is 5 K.

66) $\sum S^{\circ}_{P} - \sum S^{\circ}_{R}$
[(2 × 206.3) – (130 + 116.7)]
= 165.9 J K^{-1} mol^{-1}

70) CH≡CH 

76)

Stability of Alkene ∝ Number of α-H

77)

Addition by M.K.R

78)

Fact

79) In Wurtz reaction the simplest alkane which can be prepared is C_2H_6, and it gives product with even number of carbon atoms

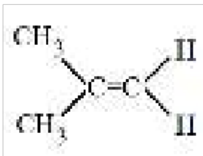
$$CH_3Br + 2Na + BrCH_3 \xrightarrow{\text{dry ether}} \underset{\text{ethane}}{CH_3 - CH_3} + 2NaBr$$

80)

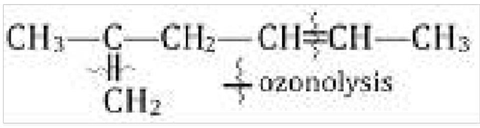
Free Radical Addtion.

81)

NCERT XI Part-II, Pg # 311

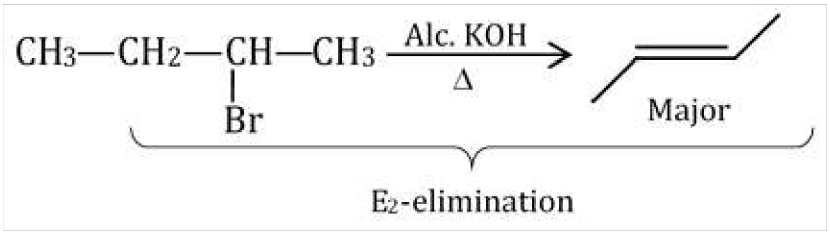


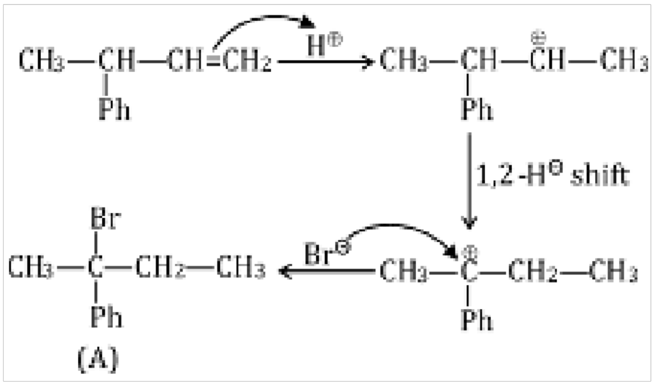
is an example of unsymmetric alkene

85) 

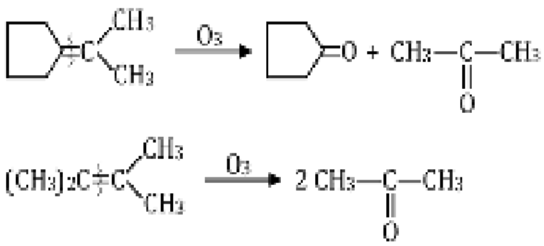
$$HCHO + CH_3-\overset{\overset{\Large O}{\|}}{C}-CH_2-CHO + CH_3-CHO$$

86) $CH_3-Cl + HCl \xrightarrow{Zn} CH_4 + ZnCl_2$

87) 

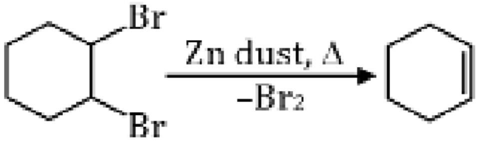
88) 

92) For two different alkyl halide Wurtz reaction is not suitable. Because three types of alkanes is formed.

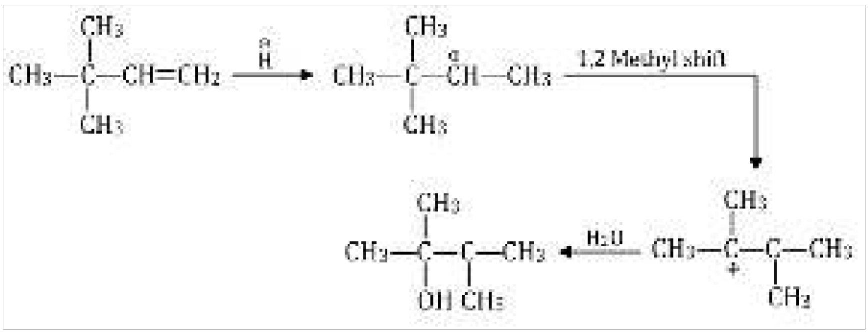


93)

94) Fact



95)



96)

97)

For one mole of an ideal gas
$C_p - C_v = R$

98)

free expansion of ideal gas
$P_{ext} = 0$
$W_{pv} = 0$
q = 0 (adiabatic process)
$\Delta E = q + w$
$\Delta E = 0$
$\Delta E = nC_{vm}\,\Delta T = 0$
$q = 0, \Delta T = 0, w = 0$

99)

$2Cl(g) \longrightarrow Cl_2(g)$
$\Delta_r S < 0$ and $\Delta_r H < 0$

100)

$2H(g) \rightarrow H_2(g)$
$\Delta n_g < 0$

BOTANY

106)

NCERT Pg. # 91

107)

NCERT XI Pg. # 89, 99

108) NCERT XI, Pg # 93

109) NCERT XI, Pg # 93, 94

110) NCERT-XI, Pg. # 89

111) NCERT XI,Pg. # 92, Para - 6.3.3

112) NCERT XI,Pg. # 89, Para - 6.2.1

113) NCERT Page # 74

114)

NCERT, Pg. # 92

116)

NCERT-XI, Pg. # 75

128) NCERT XI Pg # 26

135)

NCERT XI, Pg.# 33

137)

NCERT Pg. # 91

138)

NCERT -XI, Pg. # 34

139) NCERT XI,Pg. # 89, Para - 6.2.2

149) NCERT XI, Pg # 27

ZOOLOGY

151)

NCERT-XI, Pg. # 45
Open type circulatory system with dorsal heart.

152) NCERT XI Pg. # 48

154) NCERT Pg. # 50

157) NCERT XI, Page # 41

158) [RATIONALISED NCERT BOOK 2024-2025]
NCERT-XI, Pg # 42

159) NCERT–XI, Pg # 49

160) NCERT–XI, Pg # 53

165)

[RATIONALISED NCERT BOOK 2024-2025]
NCERT XI pg no. 42

169)

NCERT XI pg no.52

177) NCERT XI Page # 243

183) NCERT Page no. 240

184)

NCERT-XI, Pg. # 249

186) NCERT Pg. # 338

187)

NCERT Page no. 334

188)

NCERT_XI_Pg.No: 246

190)

NCERT Page no. 332

191)

NCERT Pg. No. # 330

192)

Ncert Page 333.

194) NCERT Pg.# 332

195)

NCERT Page # 247

196)

NCERT Pg. No 335